# METRO TICKET BOOKING AND RECHARGE SYSTEM USING SERVICENOW

## MINI PROJECT REPORT

**Submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology**

**Department:** Computer Science and Engineering (Artificial Intelligence & Machine Learning)

**Project Type:** Mini Project

**Platform Used:** ServiceNow

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**College:** Aditya Institute of Technology and Management (AITAM)

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# CERTIFICATE

This is to certify that the Mini Project entitled **"Metro Ticket Booking and Recharge System Using ServiceNow"** has been carried out by **Sailada Rajasekhar** in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science and Engineering (Artificial Intelligence & Machine Learning).

The work presented in this report is original and has been completed under the guidance of the project supervisor.

# DECLARATION

I hereby declare that the project report entitled **"Metro Ticket Booking and Recharge System Using ServiceNow"** is submitted by me in partial fulfillment of the requirements for the award of the Bachelor of Technology degree.

This report has not been submitted previously to any other university or institution for the award of any degree or diploma.

Sailada Rajasekhar

# ACKNOWLEDGEMENT

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# ABSTRACT

The Metro Ticket Booking and Recharge System is a ServiceNow-based application developed to simplify metro ticket booking and smart card recharge operations. Traditional metro ticket booking often requires passengers to stand in long queues at ticket counters, causing inconvenience and increasing waiting time. This project aims to digitize the entire booking process through an online service catalog.

The application provides users with two major services: booking QR-based metro tickets and recharging metro smart cards. Users can select their journey details, choose payment methods, and submit booking requests through an interactive Service Catalog interface.

Workflow automation has been implemented using ServiceNow Flow Designer. Whenever a user submits a request, the workflow automatically processes the request and stores recharge information in a custom Metro Database table. The system also generates a QR Ticket page displaying passenger information, journey details, fare amount, and ticket status.

The project demonstrates the practical implementation of ServiceNow technologies including Service Catalog, Catalog Variables, Custom Tables, Flow Designer, UI Pages, Jelly Scripting, Client Scripts, and Form Design.

This application reduces manual work, improves service efficiency, and provides a digital solution for metro ticket management.

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# CHAPTER 1

# INTRODUCTION

Urban transportation systems are expanding rapidly due to increasing population and urbanization. Metro rail systems have become one of the most preferred modes of transportation because they provide fast, reliable, and economical travel.

Despite technological improvements, many passengers still spend valuable time waiting at ticket counters for purchasing tickets or recharging metro smart cards. Manual ticket booking systems increase operational workload and often lead to long queues during peak hours.

Digital ticket booking systems eliminate these problems by allowing passengers to perform all operations online. ServiceNow provides an excellent platform for building workflow-driven applications with minimal coding effort.

This project focuses on developing a Metro Ticket Booking and Recharge System using the ServiceNow platform. The application allows passengers to:

- Book metro tickets online
- Recharge metro smart cards
- Select journey details
- Choose payment methods
- Generate request records
- Display QR tickets
- Store recharge information in a custom database

The project demonstrates how ServiceNow's enterprise workflow capabilities can be applied to build real-world transportation management solutions.

# CHAPTER 2

# PROBLEM STATEMENT

Traditional metro ticket booking systems face several challenges that reduce operational efficiency and customer satisfaction.

The major problems include:

• Long queues at metro stations.

• Manual ticket issuing process.

• Time-consuming smart card recharge.

• Increased workload for station staff.

• Lack of centralized digital records.

• Difficulty in tracking transactions.

• Delays during peak hours.

• Limited automation.

Passengers often experience inconvenience due to manual processes, while metro authorities face challenges in maintaining transaction records and managing customer requests efficiently.

The project aims to solve these problems by providing an automated ticket booking and recharge system using the ServiceNow platform.

# CHAPTER 3

# OBJECTIVES

The main objectives of the project are:

- To develop an online metro ticket booking system.
- To automate metro smart card recharge requests.
- To reduce passenger waiting time.
- To automate request processing using Flow Designer.
- To maintain recharge details in a custom database.
- To generate QR ticket pages.
- To improve customer convenience.
- To reduce manual intervention.
- To demonstrate workflow automation using ServiceNow.
- To provide a scalable enterprise application.

# CHAPTER 4

# EXISTING SYSTEM

The traditional metro ticket booking system mainly depends on manual operations.

Passengers must visit metro stations and wait in queues to purchase tickets or recharge their metro smart cards.

### Disadvantages

- Time-consuming
- Long waiting queues
- Manual record maintenance
- Human errors
- Limited accessibility
- No automation
- Difficult transaction tracking
- Higher operational cost

# CHAPTER 5

# PROPOSED SYSTEM

The proposed Metro Ticket Booking and Recharge System automates the complete booking process using ServiceNow.

Passengers can access the Service Catalog and choose one of the following services:

- Book QR Ticket
- Recharge Metro Smart Card

The application collects journey information, validates mandatory fields, processes the request using Flow Designer, and generates a request record.

Recharge requests are stored in the custom Metro Database table for future reference.

The QR Ticket page displays journey information and ticket details after successful booking.

The proposed system provides:

- Automation
- Faster processing
- Better user experience
- Digital record management
- Reduced manual work
- Workflow-based request processing
- Enterprise-level implementation

# CHAPTER 6

# SOFTWARE REQUIREMENTS

## 6.1 Hardware Requirements

| Component | Specification |
|---|---|
| Processor | Intel Core i3 / AMD Ryzen 3 or Above |
| RAM | 4 GB Minimum (8 GB Recommended) |
| Storage | 20 GB Free Disk Space |
| Internet | Stable Broadband Connection |
| Browser | Google Chrome / Microsoft Edge |
| Display | 1366 × 768 Resolution or Higher |

## 6.2 Software Requirements

| Software | Purpose |
|---|---|
| ServiceNow Personal Developer Instance | Application Development |
| Flow Designer | Workflow Automation |
| Service Catalog | User Request Interface |
| Custom Tables | Data Storage |
| UI Pages | QR Ticket Generation |
| Jelly Scripting | Dynamic UI Development |
| HTML | User Interface |
| CSS | Styling |
| JavaScript | Client-side Validation |
| Google Chrome | Testing |

## 6.3 Technologies Used

The following technologies were used throughout the project.

### ServiceNow

ServiceNow is a cloud-based enterprise platform used for workflow automation and application development. It provides low-code development tools that simplify application creation.

### Flow Designer

Flow Designer automates the business workflow. It triggers actions automatically whenever a catalog request is submitted.

### Service Catalog

The Service Catalog provides a user-friendly interface where passengers can request metro services.

### Custom Table

A custom Metro Database table stores recharge information entered by users.

### Jelly

Jelly scripting is used for creating the Metro QR Ticket page.

### HTML and CSS

Used to design the QR ticket interface.

### JavaScript

Used for client-side validation and dynamic field visibility.

# CHAPTER 7

# SYSTEM ARCHITECTURE

The Metro Ticket Booking System follows a workflow-based architecture provided by ServiceNow.

# Architecture Description

The user opens the Service Catalog and selects the Metro Ticket Booking service.

↓

The passenger enters journey details or recharge details.

↓

Client scripts validate the user input.

↓

The request is submitted.

↓

Service Catalog creates a Requested Item (RITM).

↓

Flow Designer is automatically triggered.

↓

Flow processes the request.

↓

Recharge details are stored in the Metro Database table.

↓

QR Ticket UI Page is generated.

↓

Passenger receives booking confirmation.

## Architecture Components

### User

The passenger accessing the Service Catalog.

### Service Catalog

Displays the booking and recharge forms.

### Catalog Variables

Collect passenger information.

### Flow Designer

Processes submitted requests automatically.

### Metro Database

Stores recharge records.

### UI Page

Displays the generated Metro QR Ticket.

# CHAPTER 8

# MODULES

The project consists of six major modules.

## Module 1

### User Interface Module

This module provides the Service Catalog interface where users can submit booking requests.

Features:

• Metro Ticket Booking

• Metro Recharge

• Mandatory Field Validation

• Dynamic Form Display

# Module 2

## Service Catalog Module

The Service Catalog allows users to select metro services.

Catalog Item:

Book A Metro Ticket

Options:

• Book QR Ticket

• Recharge Metro Card

# Module 3

## Flow Automation Module

Flow Designer automates the backend process.

Functions:

• Detects new requests

• Creates recharge records

• Stores request information

• Automates workflow

# Module 4

## Metro Database Module

A custom ServiceNow table stores recharge information.

Fields include:

User Details

Smart Card Number

Smart Card Name

Recharge Amount

Mode of Payment

Created Date

# Module 5

## QR Ticket Module

Displays ticket details after successful booking.

Information displayed:

Passenger Name

Journey

Source

Destination

Fare

Status

QR Code

# Module 6

## Testing Module

Ensures every function works correctly.

Includes:

Form Validation

Request Submission

Database Storage

QR Ticket Display

# CHAPTER 9

# DATABASE DESIGN

A custom table named

**Metro Database**

was created.

The table stores recharge transactions.

## Fields

| Field | Type |
|---|---|
| User Details | Reference |
| Smart Card Number | String |
| Smart Card Name | String |

| | |
|---|---|
| Recharge Amount | Integer |
| Mode of Payment | Choice |
| Created | Date/Time |

## Database Workflow

Passenger submits recharge request

↓

Flow Designer triggers

↓

Create Record Action executes

↓

Record inserted into Metro Database

↓

Recharge history maintained

## Advantages of Database

Centralized records

Easy searching

Fast retrieval

Future reporting

Data consistency

# CHAPTER 10

# FLOW DESIGN

The project uses ServiceNow Flow Designer.

Trigger:

Service Catalog

Action:

Create Record

Table:

Metro Database

Workflow Process

User submits request

↓

Flow Trigger Activated

↓

Create Record Action

↓

Insert Recharge Details

↓

Flow Ends

# Flow Components

## Trigger

Service Catalog

Starts automatically after request submission.

## Create Record

Creates a new Metro Database record.

## Data Mapping

Maps catalog inputs to database fields.

Example:

Recharge Amount → Recharge Amount

Smart Card Number → Smart Card Number

Mode of Payment → Mode of Payment

User Details → User Details

## Output

Recharge data stored successfully.

# CHAPTER 11

# IMPLEMENTATION

The implementation was completed in multiple phases.

## Phase 1

Requirement Analysis

The project requirements were collected and analyzed.

Main requirements:

Metro Ticket Booking

Metro Recharge

QR Ticket

Database Storage

Workflow Automation

## Phase 2

Catalog Creation

Created

Book A Metro Ticket

under Service Catalog.

Added catalog variables.

Implemented UI policies.

Implemented Client Scripts.

## Phase 3

Database Creation

Created custom Metro Database table.

Added recharge-related fields.

Configured form layout.

## Phase 4

Workflow Development

Created Metro Recharge Flow.

Added:

Service Catalog Trigger

Create Record Action

Mapped fields

## Phase 5

QR Ticket Development

Created UI Page.

Designed using:

HTML

CSS

Jelly

Displayed:

Passenger

Source

Destination

Fare

Journey

QR Code

## Phase 6

Testing

Performed functional testing.

Verified:

Catalog

Workflow

Database

QR Ticket

Mandatory Field

# CHAPTER 12

# TESTING

## 12.1 Introduction

Testing is an important phase in software development that ensures the developed application performs according to the specified requirements. It helps identify errors, validates system functionality, and improves reliability. In this project, testing was performed on all major modules including the Service Catalog, Flow Designer, Metro Database, and QR Ticket page.

The application was tested by entering different user inputs and verifying the outputs. Successful request generation, QR ticket display, database storage, and validation of mandatory fields were observed during testing.

## 12.2 Testing Objectives

The objectives of testing were:

- To verify that users can successfully book metro tickets.
- To validate mandatory input fields.
- To verify recharge request submission.
- To ensure Flow Designer triggers correctly.
- To verify database record creation.
- To validate QR ticket generation.
- To ensure smooth navigation across modules.

## 12.3 Test Cases

### Test Case 1

**Module:** Metro Ticket Booking

**Input:** Valid journey details

**Expected Result:** Request submitted successfully.

**Actual Result:** Request generated successfully with Request Number.

**Status:** Pass

### Test Case 2

**Module:** QR Ticket Generation

**Input:** Successful booking

**Expected Result:** QR Ticket page displayed.

**Actual Result:** Metro Ticket QR page displayed successfully.

**Status:** Pass

## Test Case 3

**Module:** Mandatory Field Validation

**Input:** Selecting "None" in mandatory fields

**Expected Result:** User should not be allowed to submit the form.

**Actual Result:** ServiceNow displayed mandatory field validation message.

**Status:** Pass

## Test Case 4

**Module:** Metro Recharge

**Input:** Valid recharge amount and payment mode

**Expected Result:** Recharge request submitted.

**Actual Result:** Request generated successfully.

**Status:** Pass

## Test Case 5

**Module:** Metro Database

**Input:** Recharge Request

**Expected Result:** Record stored in Metro Database.

**Actual Result:** Manual testing verified record creation. Automatic variable mapping requires additional Flow configuration.

**Status:** Partially Passed

### Test Case 6

**Module:** Order Status

**Input:** Request Submission

**Expected Result:** Order Status page displayed.

**Actual Result:** Request Number generated successfully.

**Status:** Pass

## 12.4 Test Summary

| Test Case | Description | Status |
|---|---|---|
| TC-01 | Metro Ticket Booking | Pass |
| TC-02 | Metro Recharge | Pass |
| TC-03 | Mandatory Field Validation | Pass |
| TC-04 | Order Status Generation | Pass |

| | | |
|---|---|---|
| TC-05 | QR Ticket Display | Pass |
| TC-06 | Database Record Creation | Partially Pass |

# CHAPTER 13

# RESULTS

The Metro Ticket Booking and Recharge System was successfully developed using the ServiceNow platform.

The implemented system provides a digital solution for metro passengers by automating the ticket booking and recharge process.

The developed application successfully demonstrated:

- Online metro ticket booking.
- Metro smart card recharge.
- Service Catalog request generation.
- Workflow automation using Flow Designer.
- QR Ticket generation.
- Request tracking through Order Status.
- Custom Metro Database table creation.
- Mandatory field validation.

The project successfully reduced manual effort and demonstrated the practical use of ServiceNow for workflow automation.

Although automatic mapping of Service Catalog variables to the custom database required additional configuration, the overall workflow and application functionality operated successfully.

# CHAPTER 14

# ADVANTAGES

The developed system offers several advantages.

- Reduces manual ticket booking.
- Minimizes waiting time.
- Provides digital ticket booking.
- Supports metro card recharge.
- Easy-to-use Service Catalog interface.
- Automated workflow processing.
- Better record management.
- Improved operational efficiency.
- Scalable architecture.
- Enterprise-ready solution.
- Low-code development.
- Easy maintenance.
- Faster request processing.
- User-friendly interface.
- Demonstrates real-world ServiceNow implementation.

# CHAPTER 15

# LIMITATIONS

Although the project is functional, a few limitations remain.

- Payment gateway is not integrated.
- QR code contains static sample information.
- Email notifications are not implemented.
- SMS alerts are not available.
- Dynamic QR validation is not implemented.
- Metro station APIs are not integrated.
- Database mapping through Flow Designer requires additional configuration.
- User authentication is basic.
- Admin reporting dashboard is not implemented.

These limitations can be addressed in future enhancements.

# CHAPTER 16

# FUTURE ENHANCEMENTS

The project can be enhanced with the following features.

## Dynamic QR Generation

Generate unique QR codes for every booking.

## Online Payment Gateway

Integrate payment gateways such as UPI, Debit Card, Credit Card, and Net Banking.

## Email Notifications

Automatically send booking confirmations via email.

## SMS Notifications

Send recharge and booking confirmations to users.

## Mobile Application

Develop an Android and iOS application connected to ServiceNow.

## Metro API Integration

Retrieve live station data, schedules, and fares from metro systems.

## Real-Time QR Validation

Allow metro gate scanners to validate QR tickets.

### Dashboard

Develop an admin dashboard for monitoring bookings and recharge statistics.

### Reports

Generate monthly and yearly reports.

### Artificial Intelligence

Recommend recharge amounts and frequently used routes based on passenger history.

# CHAPTER 17

# CONCLUSION

The Metro Ticket Booking and Recharge System was successfully designed and implemented using the ServiceNow platform. The project demonstrates how enterprise workflow automation can simplify transportation services through digital technology.

The application enables passengers to book QR tickets, recharge metro smart cards, submit requests through the Service Catalog, and receive automated request processing using Flow Designer. A custom Metro Database was created to maintain recharge records, while the QR Ticket page provided a simple digital ticket representation.

The project also demonstrated the practical use of several ServiceNow technologies, including Service Catalog, Flow Designer, Custom Tables, UI Pages, Jelly Scripting, Client Scripts, and Form Design.

Overall, the project achieved its primary objectives of reducing manual work, improving efficiency, and providing a user-friendly metro ticket management system. With additional enhancements such as payment gateway integration, dynamic QR code generation, and real-time validation, the system can be extended into a full-scale enterprise metro management solution.

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# APPENDIX

## Appendix A – Modules Developed

- Service Catalog
- Catalog Variables
- UI Policies
- Client Scripts
- Metro Database Table

- Flow Designer
- Metro Recharge Flow
- QR Ticket UI Page
- Order Status Page
- Form Validation

# PROJECT OUTCOME

This mini project successfully demonstrates a practical implementation of workflow automation using ServiceNow. It integrates Service Catalog, Flow Designer, custom database tables, UI Pages, and client-side validation to provide an online Metro Ticket Booking and Recharge System. The project showcases the use of low-code development for solving real-world transportation management problems and provides a strong foundation for future enterprise-level enhancements.